

MTH422 Endsem Q4

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Q4.

Model:

$$Y_i = f(X_i) + \varepsilon_i, \quad \varepsilon_i \sim N(0, \sigma^2), \quad f \sim \text{GP}(m(\cdot), K(\cdot, \cdot)).$$

$$X = (X_1, \dots, X_n), \quad Y = (Y_1, \dots, Y_n)^\top, \quad X_* = (x_*^{(1)}, \dots, x_*^{(m)}), \quad f_* = (f(x_*^{(1)}), \dots, f(x_*^{(m)}))^\top.$$

1: Joint distribution of $[Y, f(x_*^{(1)}), \dots, f(x_*^{(m)})]$ is multivariate normal.
by GP prior

$$\begin{pmatrix} f(x_*) \\ Y \end{pmatrix} \sim N\left(\underbrace{\begin{pmatrix} m(x_*) \\ m(X) \end{pmatrix}}_{\mu}, \underbrace{\begin{pmatrix} k(x_*, x_*) & k(x_*)^\top \\ k(x_*) & K(X, X) + \sigma^2 I_n \end{pmatrix}}_{\Sigma}\right),$$

notation:

$$k(x_*) = \begin{pmatrix} K(x_*, X_1) \\ \vdots \\ K(x_*, X_n) \end{pmatrix}, \quad [K(X, X)]_{ij} = K(X_i, X_j), \quad i, j = 1, \dots, n.$$

$$[m(X)]_i = m(X_i), \quad [m(x_*)]_j = m(x_*^{(j)}),$$

2: here PPD is Conditional Distribution on Y that is also another MVN over $\mathbf{f}(\mathbf{x}^*)$. (Using MVN conditioning identity.)

If $\begin{pmatrix} A \\ B \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_A \\ \mu_B \end{pmatrix}, \begin{pmatrix} \Sigma_{AA} & \Sigma_{AB} \\ \Sigma_{BA} & \Sigma_{BB} \end{pmatrix}\right)$, then

$$A \mid B = b \sim N\left(\mu_A + \Sigma_{AB} \Sigma_{BB}^{-1} (b - \mu_B), \Sigma_{AA} - \Sigma_{AB} \Sigma_{BB}^{-1} \Sigma_{BA}\right).$$

in our question

$$A = f_*, \quad B = Y, \quad \mu_A = m(x_*), \quad \mu_B = m(X),$$

$$\Sigma_{AA} = K(x_*, x_*), \Sigma_{AB} = k(x_*)^\top, \Sigma_{BB} = K(X, X) + \sigma^2 I_n, \Sigma_{BA} = k(x_*)^\top,$$

3: Posterior mean and covariance.

$$f_* \mid Y \sim N\left(m^*(x_*), K^*(x_*, x'_*)\right),$$

using the result of conditional mean and var-cov matrix:

$$\boxed{\begin{aligned} m^*(x_*) &= m(x_*) + k(x_*)^\top [K(X, X) + \sigma^2 I_n]^{-1} (Y - m(X)), \\ K^*(x_*, x'_*) &= K(x_*, x'_*) - k(x_*)^\top [K(X, X) + \sigma^2 I_n]^{-1} k(x'_*), \end{aligned}}$$

$$k(x_*) = \begin{pmatrix} K(x_*, X_1) \\ \vdots \\ K(x_*, X_n) \end{pmatrix}, \quad [K(X, X)]_{ij} = K(X_i, X_j), \quad i, j = 1, \dots, n.$$

Matching my result with Answer key

matching my notation from answer key:

$$\mu = m(X), \quad \mu^* = m(x_*), \quad \Sigma = K(X, X), \quad \Sigma^* = k(x_*),$$

1 Mean

we can either use

1. use MVN conditioning identity to find

$$E[f \mid Y]$$

or

2. Matrix Algebra to solve final expression of mean

1.1 using MVN conditioning identity from section 2

$$E[f_* \mid Y] = \mu^* + \Sigma^* \Sigma^{-1} (E[f \mid Y] - \mu).$$

Now we find $E[f \mid Y]$ using same result of section 2 partitioned-Gaussian conditioning formula for f and Y .

$\begin{pmatrix} A \\ B \end{pmatrix} \sim N((\mu_A, \mu_B), (\Sigma_{AA}, \Sigma_{AB}; \Sigma_{BA}, \Sigma_{BB}))$, then

$$E[A \mid B] = \mu_A + \Sigma_{AB} \Sigma_{BB}^{-1} (B - \mu_B).$$

Here $A = f$, $B = Y$, so $\Sigma_{AB} = \Sigma$ and $\Sigma_{BB} = \Sigma + \sigma^2 I_n$. so

$$E[f \mid Y] = \mu + \Sigma [\Sigma + \sigma^2 I_n]^{-1} (Y - \mu).$$

Substitute back:

$$\begin{aligned} E[f_* \mid Y] &= \mu^* + \Sigma^* \Sigma^{-1} (\mu + \Sigma [\Sigma + \sigma^2 I_n]^{-1} (Y - \mu) - \mu) \\ &= \mu^* + \Sigma^* [\Sigma + \sigma^2 I_n]^{-1} (Y - \mu), \end{aligned}$$

as $\mu - \mu = 0$ and $\Sigma^{-1} \Sigma = I$. In my kernel notation this is exactly same as

$$\boxed{m^*(x_*) = m(x_*) + k(x_*)^\top [K(X, X) + \sigma^2 I_n]^{-1} (Y - m(X)).}$$

1.2 using matrix Algebra

$$\text{let } M = \sigma^{-2}I_n + \Sigma^{-1}$$

(there is a correction in the below expression of your answer key (in the photo i took), there should be $\Sigma^{-1}\mu$ instead of $\Sigma^{-1}f$. in notes it is correct).

$$\begin{aligned} E[f_* | Y] &= \mu^* + \Sigma^* \Sigma^{-1} \left(M^{-1} [\sigma^{-2}Y + \Sigma^{-1}\mu] - \mu \right) \\ &= \mu^* + \Sigma^* \Sigma^{-1} \left(M^{-1}(\sigma^{-2}Y + \Sigma^{-1}\mu) - M^{-1}M\mu \right) \quad (M\mu = \sigma^{-2}\mu + \Sigma^{-1}\mu) \\ &= \mu^* + \Sigma^* \Sigma^{-1} M^{-1} (\sigma^{-2}Y + \Sigma^{-1}\mu - (\sigma^{-2}\mu + \Sigma^{-1}\mu)) \\ &= \mu^* + \Sigma^* \Sigma^{-1} M^{-1} (\sigma^{-2}(Y - \mu)) \\ &= \mu^* + \Sigma^* \underbrace{\Sigma^{-1}M^{-1}\sigma^{-2}}_{=(\Sigma + \sigma^2 I_n)^{-1}} (Y - \mu) \\ &= \mu^* + \Sigma^* (\Sigma + \sigma^2 I_n)^{-1} (Y - \mu) \end{aligned}$$

this is because $M = \sigma^{-2}I_n + \Sigma^{-1} = \sigma^{-2}\Sigma^{-1}(\Sigma + \sigma^2 I_n)$,

$$\begin{aligned} M^{-1} &= [\sigma^{-2}\Sigma^{-1}(\Sigma + \sigma^2 I_n)]^{-1} = (\Sigma + \sigma^2 I_n)^{-1} \Sigma \sigma^2, \\ \Sigma^{-1}M^{-1}\sigma^{-2} &= \Sigma^{-1}[(\Sigma + \sigma^2 I_n)^{-1} \Sigma \sigma^2] \sigma^{-2} = (\Sigma + \sigma^2 I_n)^{-1} \underbrace{\Sigma^{-1}\Sigma}_{I_n} = (\Sigma + \sigma^2 I_n)^{-1}. \end{aligned}$$

so

$$E[f_* | Y] = \mu^* + \Sigma^* (\Sigma + \sigma^2 I_n)^{-1} (Y - \mu)$$

in my kernel notation it is exactly same as

$$= m(x_*) + k(x_*)^\top [K(X, X) + \sigma^2 I_n]^{-1} (Y - m(X)).$$

2 Covariance Matrix

similarly again using the section 2 to get conditional covariance matrix

$$Cov(f | Y) = \Sigma - \Sigma (\Sigma + \sigma^2 I_n)^{-1} \Sigma.$$

put this in

$$Cov(f_* | Y) = \Sigma^* \Sigma^{-1} Cov(f | Y) \Sigma^{-1} (\Sigma^*)^\top + \Sigma^{**} - \Sigma^* \Sigma^{-1} (\Sigma^*)^\top.$$

$$\begin{aligned} Cov(f_* | Y) &= \Sigma^* \Sigma^{-1} \left[\Sigma - \Sigma (\Sigma + \sigma^2 I_n)^{-1} \Sigma \right] \Sigma^{-1} (\Sigma^*)^T + \Sigma^{**} - \Sigma^* \Sigma^{-1} (\Sigma^*)^T \\ &= \Sigma^* \Sigma^{-1} \Sigma \Sigma^{-1} (\Sigma^*)^T - \Sigma^* \Sigma^{-1} \Sigma (\Sigma + \sigma^2 I_n)^{-1} \Sigma \Sigma^{-1} (\Sigma^*)^T + \Sigma^{**} - \Sigma^* \Sigma^{-1} (\Sigma^*)^T \\ &= \Sigma^* \Sigma^{-1} (\Sigma^*)^T - \Sigma^* (\Sigma + \sigma^2 I_n)^{-1} (\Sigma^*)^T + \Sigma^{**} - \Sigma^* \Sigma^{-1} (\Sigma^*)^T \\ &= \Sigma^{**} - \Sigma^* (\Sigma + \sigma^2 I_n)^{-1} (\Sigma^*)^T. \end{aligned}$$

in my kernel notation it is same as

$$\boxed{K^*(x_*, x'_*) = K(x_*, x'_*) - k(x_*)^\top [K(X, X) + \sigma^2 I_n]^{-1} k(x'_*)}.$$